26. A class starts at 11 : 00 am and lasts till 2 : 27 pm. Four periods of equal duration are held during this interval. After every period, a rest of 5 minutes is given to the students. The exact duration of each period is:

(a) 48 minutes (b) 50 minutes
(c) 51 minutes (d) 53 minutes

Ans. (a) : Let the time of each period = x min
$\because$ Number of period = 4
$\therefore$ Number of interval = 3
According to question,
$4 \times x + 3 \times 5 = (2:27\text{p.m-}11\text{ a.m.})$
$4x + 15 = 3$ hr 27 min
$4x + 15 = 207$ min
$4x = 207 - 15$
$4x = 192$
$x = 48$ min.
Each time A will go with his friends and last he will go alone.

27. Four friends A, B, C and D need to cross a bridge. A maximum of two persons can cross it at a time. It is night and they just have one lamp. Persons that cross the bridge must carry the lamp to find the way. A pair must walk together at a speed of slower person. After crossing the bridge, the person having faster speed in the pair will return with the lamp each time to accompany another person in the group. Finally, the lamp has to be returned at the original place and the person who returns the lamp has to cross the bridge, again without lamp. To cross the bridge, the time taken by them is as follows : A : 1 minute , B : 2 minutes C : 7 minutes and D : 10 minutes. What is the total minimum time required by all the friends to cross the bridge?

(a) 23 minute (b) 22 minute
(c) 21 minute (d) 20 minute

Ans. (a) :
The total time taken by A, B, C and D $\Rightarrow$
D (10 + 1) min + C (7 + 1) min + B (2 + 1) min + A (1) min
$\Rightarrow$ (11 + 8 + 3 + 1) min
= 23 min

28. 30g of sugar was mixed in 180 ml water in a vessel A, 40 g of sugar Was mixed in 280 ml of waterin vessel B and 20 g of sugar was mixed in 100 ml of water in vessel C. The solution in vessel B is

(a) sweeter than that in C
(b) sweeter than that in A
(c) as sweet as that in C
(d) less sweet than that in C

Ans. (d) : Sugar in per ml water $\Rightarrow$

$$A = \frac{30}{180} \Rightarrow \frac{1}{6}$$

$$B = \frac{40}{280} \Rightarrow \frac{1}{7}$$

$$C = \frac{20}{100} = \frac{1}{5}$$

Hence,

$$\frac{1}{5} > \frac{1}{6} > \frac{1}{7}$$

C > A > B
↓ ↓ ↓
Sweetest Sweeter Sweet

Hence, the solution in vessel B is less sweet than that in C.

29. In aid of charity, every student in a class contributes as many rupees as the number of students in that class. With the additional contribution of Rs. 2 by one student only the total collection is Rs. 443. Then how many students are there in the class?

(a) 12 (b) 21
(c) 43 (d) 45

Ans. (b) : The number of student in the class

$$= \sqrt{443 - 2}$$
$$= \sqrt{441}$$
$$= 21$$

30. Anita's mathematics test had 70 problems carrying equal marks i.e., 10 arithmetic, 30 algebra and 30 geometry. Although she answered 70% of the arithmetic, 40% of the algebra and 60% of the geometry problems correctly, she did not pass the test because she got less then 60% marks. The number of more questions she would have to answer correctly to earn a 60% Passing marks is .

(a) 1 (b) 5
(c) 7 (d) 9

Ans. (b) :
Number of questions correct by Anita

$$\Rightarrow 10 \times \frac{70}{100} + 30 \times \frac{40}{100} + 30 \times \frac{60}{100}$$
$$= 7 + 12 + 18$$
$$= 37$$

The number of correct questions required to pass the exam $= 70 \times \frac{60}{100} = 42$

$\therefore$ The numberof more questions to earn passing mark
$42 - 37 \Rightarrow 5$

31. In the class, there are 18 very tall boys. If these constitute three –fourths of the boys and the total number of boys is two –thirds of the total number of students in the class, what is the number of girls in the class ?

(a) 6 (b) 12
(c) 18 (d) 21

Ans. (b) : Let the number of boys in the class and the number of girls in the class
According to question,

$$\frac{3x}{4} = 18 \Rightarrow x = \frac{18 \times 4}{3} = 24$$

Again,

$$24 = (24+y) \times \frac{2}{3}$$
$$2(24+y) = 24 \times 3$$
$$24 + y = \frac{24 \times 3}{2}$$
$$24 + y = 36$$
$$y = 36 - 24 = 12$$

The number of girls = 12

32. Consider the following statements :

1. Either A and B are of the same age or A is older than B
2. Either C and D are the same age or D is older than C
3. B is older than C

Which of the following conclusions can be drawn from the above statements ?

(a) A is older than B
(b) B and D are of the same age
(c) D is older than C
(d) A is older than C

Ans. (d) :

$A \geq B$ .....(i)
$D \geq C$ .....(ii)
$B > C$ .....(iii)

from eq (i), (ii) & (iii)
Hence A is older than C.

33. The monthly average salary paid to all the employees of a company was Rs. 5000. The monthly average salary paid to male and female employees was Rs. 5200 and Rs. 4200 respectively. The percentage of males employed in the company is

(a) 75% (b) 80%
(c) 85% (d) 90%

Ans. (b) :
Let
Male = x
Female = y
According to question
$x \times 5200 + y \times 4200 = (x+y)\ 5000$
$5200x + 4200\ y = 5000x + 5000y$
$200\ x = 800\ y$
$x = 4\ y$
The percentage of male employed of company

$$= \frac{x}{x+y} \times 100$$

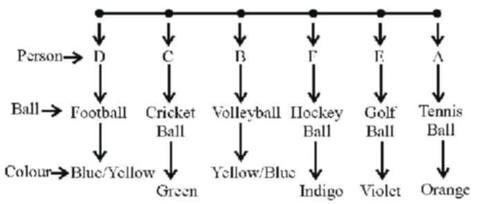
$\{\because x = y\}$

$$= \frac{4y}{4y+y} \times 100 = \frac{4y}{5y} \times 100 = 80\%$$

Direction for the following 3 (three) items: Consider the given formation and answer the three items that follow.

Six boxes A,B, C, D, E and F have been painted with six different colours viz. violet, Indigo, blue, green, yellow and orange and arranged from left to right (not necessarily either kept or painted with the colours in the same order). Each box contains a ball of any one of the following six games: cricket, hockey, tennis, golf, football and volleyball (not necessarily in the same order). The golf ball is in violet box and is not in the box D. The box A which contains tennis ball is orange in colour and is at the extreme right. The hockey ball is neither in box D nor in box E. The box C having cricket ball is painted blue nor in the box painted yellow. The box C is fifth from right and next to box B. The box B contains volleyball. The box containing the hockey ball is between the boxes containing golf ball and volley ball.

Solution for 34 to 36



34. Which one of the following boxes contains the golf ball ?

(a) F
(b) E
(c) D
(d) None of the above

Ans. (b) : E box contains the golf ball.

35. Which of the following statements is/are correct ?

(a) D is painted yellow
(b) F is painted indigo
(c) B is painted blue
(d) All of the above

Ans. (b) : 'F is painted indigo" is correct.

36. The football is in the box of which colour ?

(a) Yellow
(b) Indigo
(c) Cannot be determined as data are inadequate
(d) Blue

Ans. (c) : The football is in either yellow or blue colour.
Hence, colour can not be determined or data are inadequate.

37. Two number X and Y are respectively 20% an 28% less than a third number Z. By what percentage is the number Y less han the number X ?

(a) 12% (b) 10%
(c) 9% (d) 8%